

PATHOLOGY

Two-dimensional echocardiography is widely used to assess ventricular function, the anatomy of the proximal coronary arteries, and the presence of pericardial effusion. According to American Heart Association guidelines, patients with uncomplicated Kawasaki syndrome (KS) should undergo echocardiographic evaluation at the time of diagnosis, at 2 weeks, and again at 6 to 8 weeks. In selected cases, magnetic resonance imaging of the proximal coronary artery segments may provide additional information. Cardiac stress testing can detect reversible ischemia in children with coronary artery aneurysms. Coronary arteriography is typically reserved for patients with persistent echocardiographic abnormalities or symptoms of myocardial ischemia. It is valuable for visualizing coronary artery stenoses and assessing the extent of collateral vessel formation.

Extensive gross and microscopic pathological studies have been conducted in autopsies of KS patients who died during various phases of the illness. The overall case fatality rate for KS is approximately 0.4%, with most deaths attributable to cardiovascular complications. The pathological course of KS is divided into four stages based on the duration of illness at the time of death.

Stage 1 (0 to 9 days after onset)

- Characterized by acute vasculitis and perivascularitis of small blood vessels.
- Microscopic findings include endothelial activation and endothelial cell damage.
- Infiltration of neutrophils and mononuclear cells (activated T cells and monocytes/macrophages) is present.
- Expression of MHC class II antigens on coronary arterial endothelium is observed in KS but not in normal controls.
- Cytokines such as interferon- γ (a potent inducer of MHC class II), IL-1, and TNF- α play key roles in inducing leukocyte adhesion molecules.

- These findings suggest that cytokine-mediated endothelial antigen expression contributes significantly to disease pathogenesis.
- In larger arteries, inflammation initially involves the intima and vasa vasorum, with relative sparing of the media.
- Additional findings may include pericarditis, myocarditis, inflammation of the atrioventricular conduction system, and endocarditis with valvulitis.

Stage 2 (12 to 25 days after onset)

- Marked infiltration of inflammatory cells extends into the media of coronary and other medium-sized arteries.
- Aneurysm formation begins, often accompanied by thrombosis.
- Destruction of the vessel wall is thought to result from the release of proteolytic enzymes and inflammatory mediators into the arterial media.

Stage 3 (28 to 31 days after onset)

- Acute inflammation begins to subside.
- Granulation tissue forms in the coronary arteries.

Stage 4 (40 days to 4 years after onset)

- Chronic phase characterized by scar formation, organization of thrombi, intimal proliferation, and neoangiogenesis.
- These processes may continue for years and can lead to severe stenosis of the coronary arteries and other medium-sized vessels.

It is important to note that while coronary arteries are consistently involved in KS, virtually any medium- or large-sized blood vessel can be affected, as demonstrated in autopsy studies. Aneurysms have been reported in the femoral, iliac, renal, axillary, and brachial arteries. KS is a systemic vasculitis primarily affecting small- and medium-sized blood vessels. However, cutaneous vasculitis is not a prominent clinical feature.